BIS108 BUSINESS REQUIREMENTS ANALYSIS | ASSESSMENT 4

Hartwell Primary School Sustainable Travel System

Boroondara Schools Project Prototype, CASE Tool Analysis & Usability Verification

Lecturer & Client: Akrati Jain | Member A: Sofia Mironenko (UC-01–UC-03) | Live Prototype: A4_Group_Prototype_Offline.html

ITEM 1: PROTOTYPING APPROACH SELECTION

Prototyping Approach Evaluation Matrix

Evaluation Criterion	Evolutionary (Selected)	Throwaway (Rejected)	Horizontal (Rejected)
Adaptiveness to Change	5/5: Modular updates	2/5: Rebuilt from zero	3/5: Surface labels only
End-to-End User Goal	5/5: Full task journeys	3/5: Isolated fragments	1/5: Superficial shells
Client Sign-Off Utility	5/5: Realistic controls	2/5: Low-fidelity sketch	2/5: Hides workflow logic
Iterative Alignment	5/5: Continuous baseline	3/5: Discovery spikes only	2/5: Premature breadth
Total Weighted Score	**20 / 20 (Optimal)**	**10 / 20 (Disqualified)**	**8 / 20 (Disqualified)**

ITEM 1: RATIONALE & GOVERNANCE (TILLEY, 2025; WIEGERS & HOKANSON, 2023)

Selection Justification & Risk Mitigation

Why Evolutionary Won

• Iterative Baseline: Refines high-priority safety workflows continuously.

• Throwaway Disqualified: Rebuilding wastes the $40k–$50k budget.

• Horizontal Disqualified: Shallow menus fail 500m hub check-ins.

• Goal Validation: Verifies end-to-end non-smartphone child tracking.

Identified Risk & Mitigation

• Core Risk: Scope creep from polished UI appearance.

• False Readiness: Mistaking front-end simulation for live backend.

• SRS Anchoring: Enforce strict SRS Use Case boundaries (UC-01–09).

• Change Control: Restrict edits to formal requests (CR-01).

CASE Tool Investigation & Decisive Criterion

CASE Tool Package	1. Cost	2. Learning Curve	3. Classroom Run	4. Adaptiveness	Total
Figma + Browser HTML (Selected)	**5/5 Free Edu**	**5/5 Intuitive UI**	**5/5 Zero-Install Browser**	**5/5 Global Tokens (<10s)**	**20 / 20**
Balsamiq Wireframes (Rejected)	3/5 30d Trial	5/5 Simple Sketch	3/5 Fails Colour Rubric	2/5 Manual Screen Edits	**13 / 20**
Axure RP 10 (Rejected)	2/5 $25–$49/m	2/5 Steep Logic	3/5 Heavy Local Install	4/5 Slow Team Sync	**11 / 20**

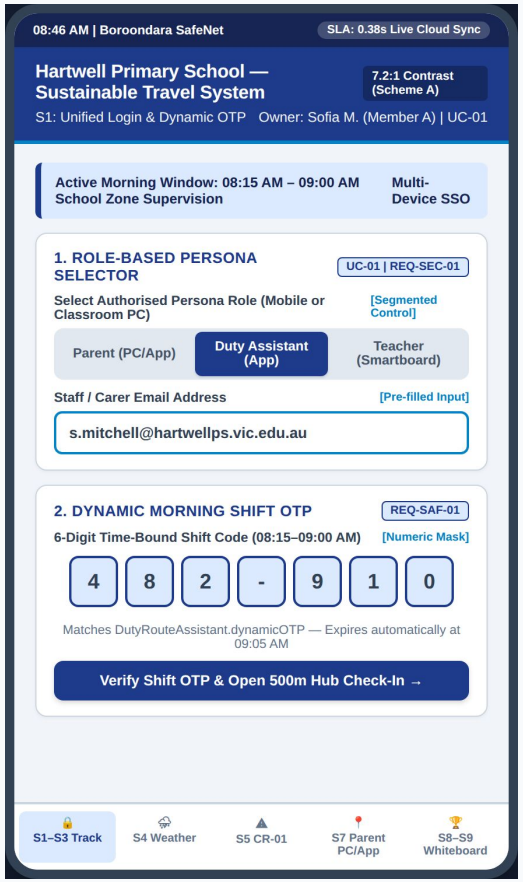
Decisive Criterion: Adaptiveness to Change & Zero-Install Classroom Operability (Global colour swap across 9 screens in <10s).

ITEM 3: CLIENT CONSULTATION & COLOUR SCHEMES

Client Colour Schemes & Change Request CR-01

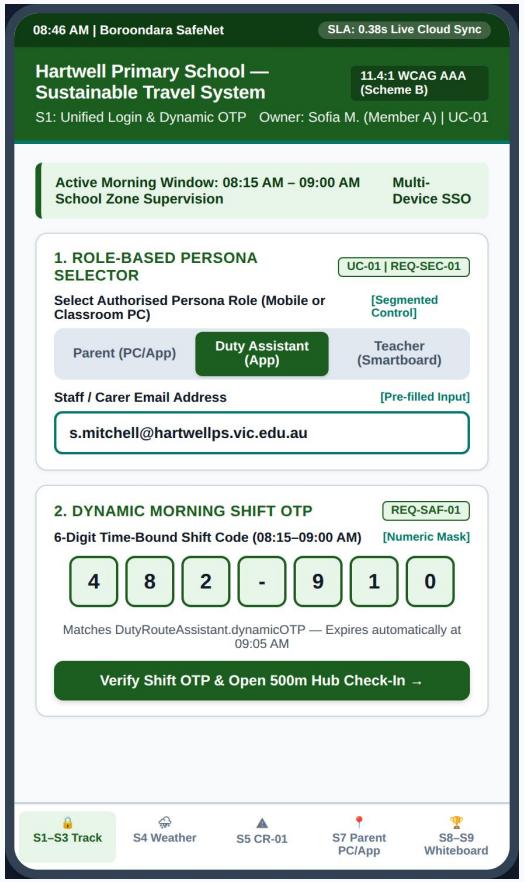
Scheme A: Municipal Navy (Rejected — 7.2:1 Contrast)

Corporate feel lacked environmental identity.



Scheme B: Eco-Safe Green (Client Selected — 11.4:1 AAA)

Outdoor daylight contrast + Client Change CR-01 on S5.



ITEM 3 & 5: COMPLETE SCREENS TRACEABILITY & ATTRIBUTION

9-Screen SRS Traceability & Group Attribution

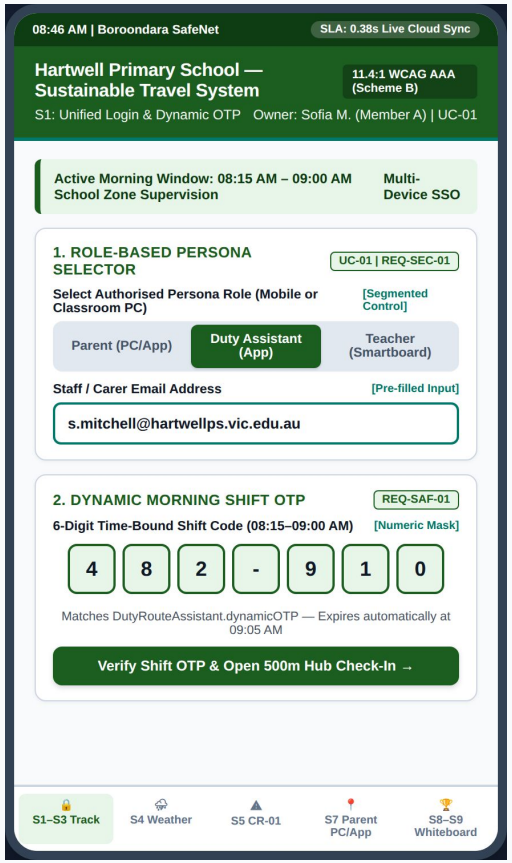
Subsystem Owner	Attributed Screens	SRS Use Cases & Req IDs	Matched UI Controls Implemented
Member A (Sofia Mironenko)	S1 Login, S2 500m Check-In, S3 9AM SOS & Summary S7 Parent Map, S8 Privacy & Leaderboard, S9 Council Export	UC-01 to UC-03 | REQ-SEC-01, REQ-TRK-01/02, REQ-SAF-01/02 UC-07 to UC-09 | REQ-PAR-01, REQ-ETH-01, REQ-ACC-01	Segmented Role, 6-Digit OTP, Calendar Date Picker, 2-Step SOS Modal Sibling Dropdown, Time-Bounded GPS Window, Locked PII Toggle, 125% Slider
Member B	S4 BOM Feed, S5 CR-01 Override, S6 Hub Roll Call & KPIs	UC-04 to UC-06 | REQ-ENV-01/02, REQ-TRK-03, REQ-HAZ-01/02	Corridor Dropdown, Constrained CR-01 Detour Dropdown, 1:1 Check, KPI Gauges

MEMBER A WALKTHROUGH: SOFIA MIRONENKO (UC-01 TO UC-03)

Duty Assistant Safety & 9:00 AM Escalation Goal

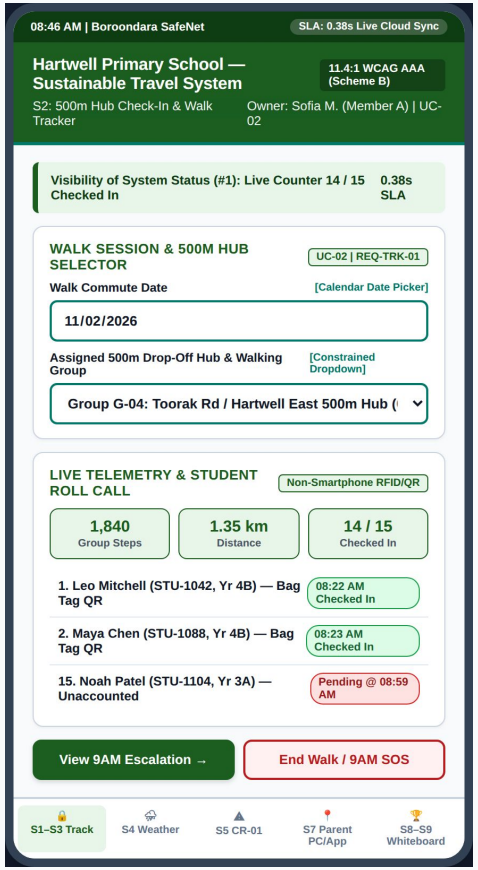
S1: Dynamic OTP Login

Segmented role + 8:15–9:00 AM OTP



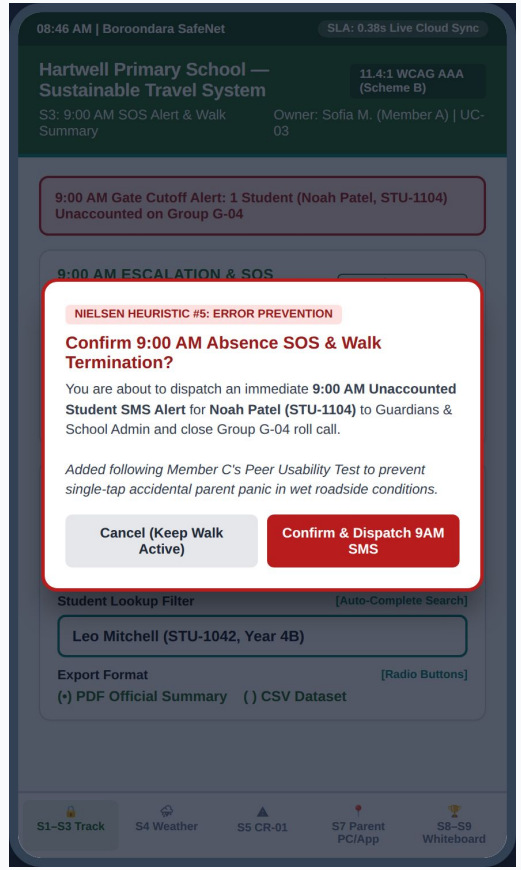
S2: 500m Hub Check-In

Date picker + Toorak Rd 500m dropdown



S3: 9:00 AM SOS Alert

2-step confirmation modal prevents panic

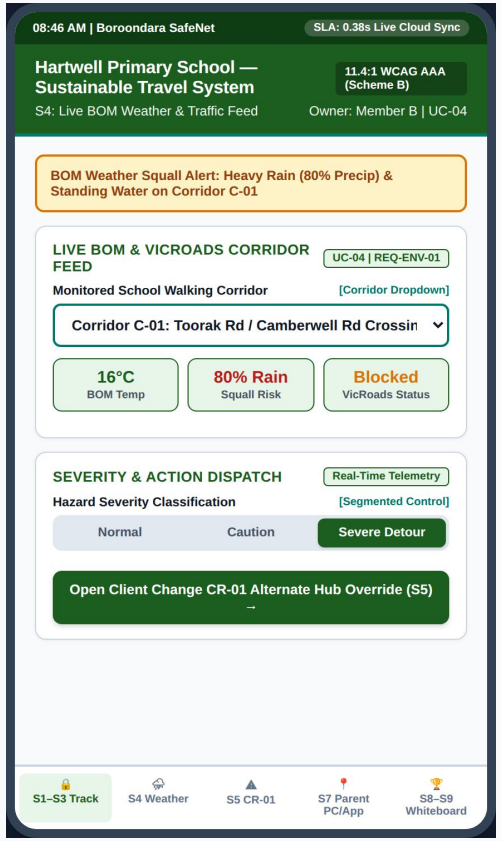


Limitation & Iteration 2 Fix: Add offline local cache queue with 3-minute grace check before firing 9:00 AM SMS.

Weather Hazard Re-Routing & Congestion KPIs

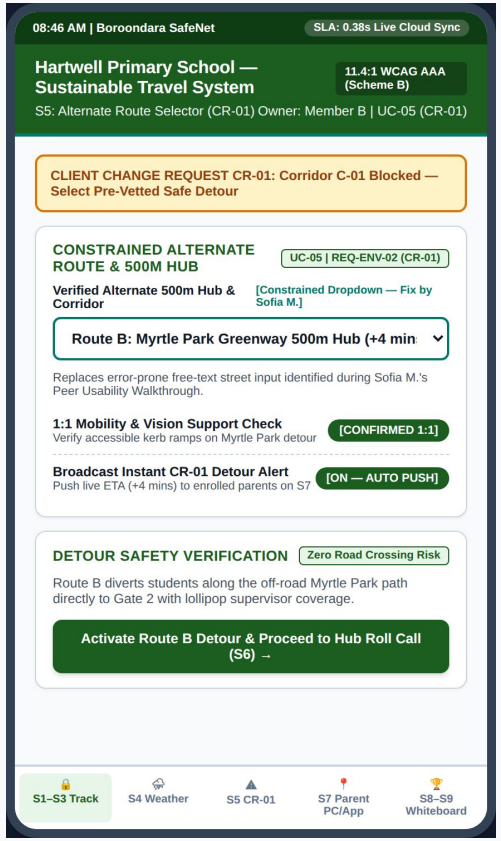
S4: BOM & VicRoads Feed

Detects heavy rain squall on Corridor C-01



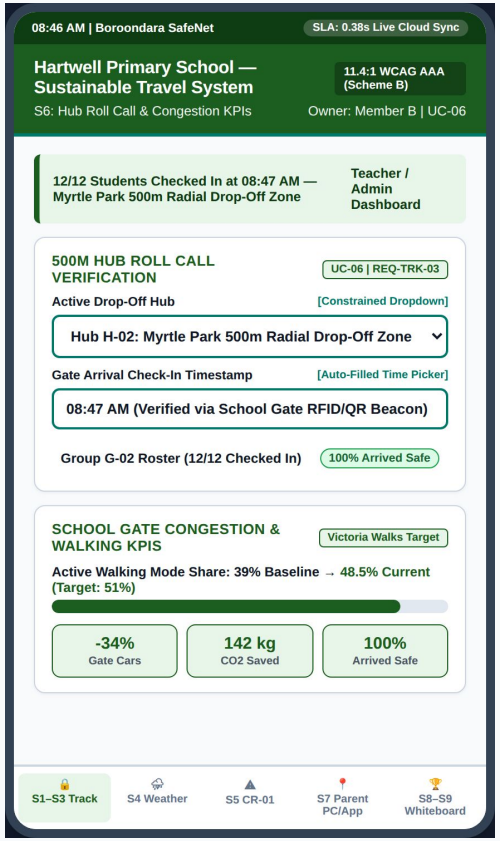
S5: Admin Hub Override (CR-01)

Diverts group to Myrtle Park 500m Hub



S6: Roll Call & Congestion KPIs

12/12 Checked In + 39% to 51% target

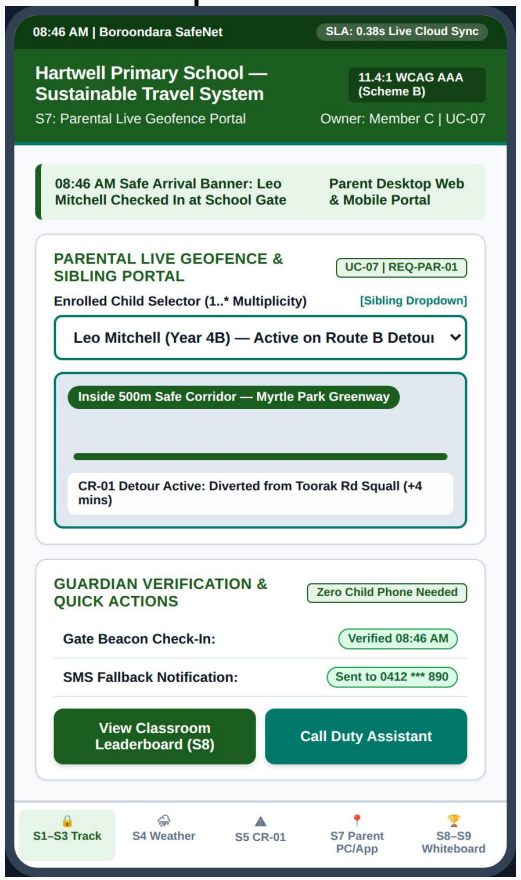


Limitation & Iteration 2 Fix: Add offline local sync queue indicator for park drop-off hubs with weak 4G reception.

Guardian Visibility, Gamification & Privacy Compliance

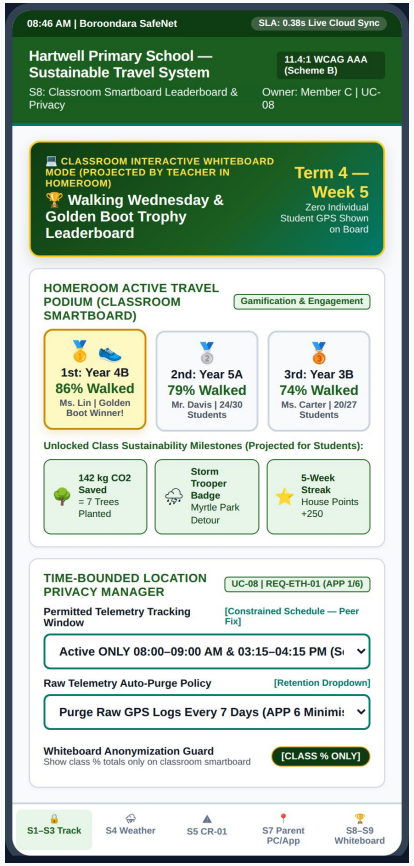
S7: Parent Arrival Portal

08:46 AM safe arrival banner without child phone



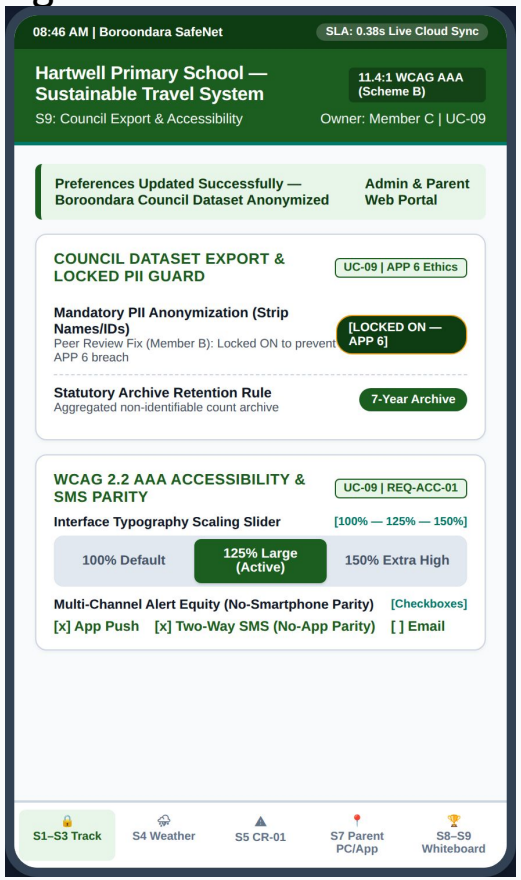
S8: Privacy & Leaderboard

08:00–09:00 AM GPS window + Year 4B 86%



S9: Council Export & Access

Locked PII anonymization + 125% scaling



Limitation & Iteration 2 Fix: Embed live fallback SMS preview card for parents without smartphone data.

ITEM 4: INTERFACE USABILITY & CLASSROOM EXECUTION

Nielsen Usability Heuristics & Run Guide

3 Named Heuristics (Nielsen, 1994)

1. Visibility of System Status (#1): Live 14/15 check-in counter (S2), 0.38s SLA pill, 08:46 AM arrival banner (S7).

2. Error Prevention (#5): Constrained 500m hub dropdowns (S2, S5), two-step SOS modal (S3), locked PII toggle (S9).

3. Consistency & Standards (#4): Persistent 9-screen menu, uniform field labels, 11.4:1 WCAG AAA contrast.

Zero-Install Classroom Run Steps

1. Open Browser: Launch Chrome, Edge, or Firefox on classroom PC.

2. Open File/Link: Double-click A4_Group_Prototype_Offline.html or open Figma URL.

3. Test Colour Schemes: Click top-right Scheme A/B switcher (<10s swap).

4. Execute Goals: Use left menu (S1–S9) or bottom 5-tab bar for end-to-end tasks.

Peer Usability Testing Log & References

Reviewer	Target Screen Tested	Usability / Ethical Defect Found	Fix Implemented in Final Build
Sofia M. (Member A)	S5 Override (Member B)	Free-text detour input risked typos in rain Single-tap SOS button risked false parent panic	Constrained 500m Hub dropdown + 1:1 check Two-step Error Prevention confirmation modal
Member B (kim on computer)	S8 & S9 Export (Member C)	Unchecking PII anonymization & 24/7 GPS risked APP 6 breach	Locked-ON PII toggle + 08:00–09:00 AM window

References (APA 7th): ISO (2019, ISO 9241-210); Nielsen, J. (1994, Usability engineering); OAIC (2021, APP guidelines); Tilley, S. (2025, Systems analysis and design, 13th ed.); Victoria Walks (2010); Wiegers & Hokanson (2023).

WEEK 13 LIVE DEMONSTRATION

Questions & Live Prototype Walkthrough

All 9 screens (S1–S9) and the live Client Colour Scheme Switcher (Scheme A vs. Scheme B) are ready for inspection.